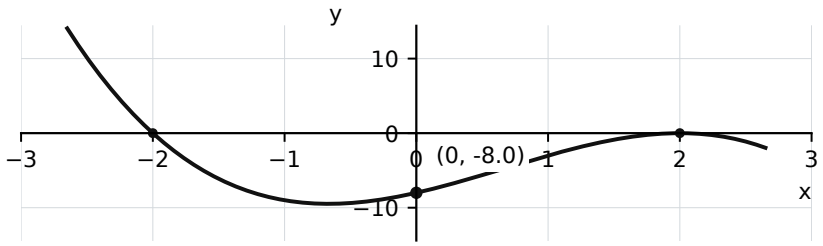
MATH 101 **75395**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.  	1. 10 pts
2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+9} - 2$	2. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

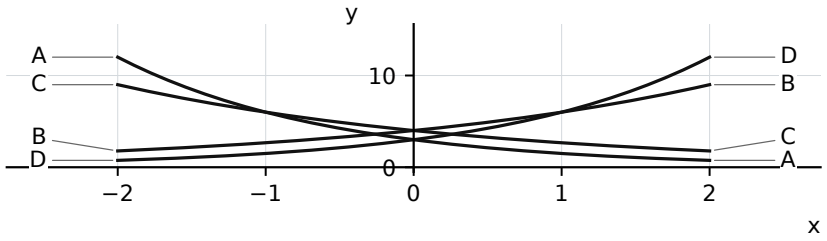
3. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 4(\frac{2}{3})^x$

$(b)\quad 3(\frac{1}{2})^x$

$(c)\quad 3(2)^x$

$(d)\quad 4(\frac{3}{2})^x$



3. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

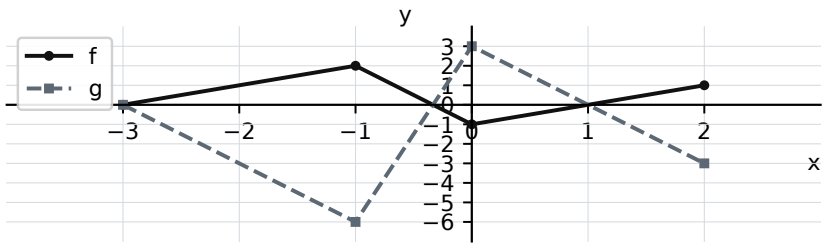
4. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x) - 3$

$B.\quad g(x) = -3f(x)$

$C.\quad g(x) = 3f(x)$

$D.\quad g(x) = f(-3x)$



4.

10 pts

Test continues on the next sheet.

MATH 101 **75395**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Solve exactly. $\sqrt{x+2} = x-2$	6. 10 pts
7. Which pair satisfies h(x) = f(g(x))? $h(x) = (3x+3)^2 + 4$ $A.\quad f(x) = x^2 + 4,\ g(x) = 3x+3$ $B.\quad f(x) = x^2 + 3,\ g(x) = 3x+4$ $C.\quad f(x) = 3x^2 + 4,\ g(x) = x+3$ $D.\quad f(x) = 3x+3,\ g(x) = x^2+4$	7. 10 pts

Test continues on the other side of the sheet.

MATH 101 **75395**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-2) + 1$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
9. Evaluate exactly. Here log means base 10. (a) $\log(0.8) + \log(1250)$ (b) $\frac{\log_6(27)}{\log_6(9)}$	9. (a) 5 pts (b) 5 pts
10. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	10. 10 pts

75395

TEST PAGE 1

1.

$-1(x+2)(x-2)^2$

10 pts

2. (a)

$[-9, \infty)$ 2 pts

(b)

$[-2, \infty)$ 2 pts

(c)

$(-5, 0)$ 1 pts

(d)

$(0, 1)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -2$ 1 pts

TEST PAGE 2

3. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

4.

B

10 pts

TEST PAGE 3

5. (a)

$\mathbb{R} \setminus \{-2, 5\}$ 2 pts

(b)

$(-2, -\frac{1}{7})$ 2 pts

(c)

$x = 5$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-3, 0)$ 2 pts

(f)

$(0, -\frac{3}{5})$ 2 pts

6.

$x = \frac{\sqrt{17}}{2} + \frac{5}{2}$ 10 pts

7.

A

10 pts

75395

TEST PAGE 4

8. (a)

$(2, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 2$ 1 pts

(d)

$(\frac{7}{3}, 0)$ 1 pts

(e)

$x \to 2^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

9. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

10.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 10 pts

MATH 101 **79298**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-2)+1$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = \sqrt{x+16}-2$	2. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
3. Evaluate exactly. Here log means base 10. (a) $\log(0.05)+\log(200000)$ (b) $\frac{\log_6(27)}{\log_6(9)}$	3. (a) 5 pts (b) 5 pts

Test continues on the other side of the sheet.

MATH 101 **79298**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

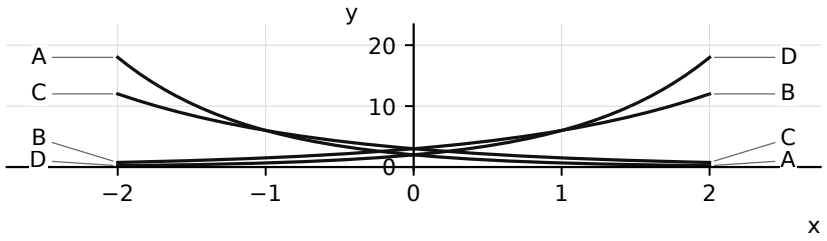
4. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(\frac{1}{2})^x$

$(b)\quad 2(\frac{1}{3})^x$

$(c)\quad 2(3)^x$

$(d)\quad 3(2)^x$



4. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

5. Which pair satisfies h(x) = f(g(x))?

$h(x) = (3x + 3)^2 + 4$

$A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 3$

$B.\quad f(x) = x^2 + 3,\ g(x) = 3x + 4$

$C.\quad f(x) = 3x^2 + 4,\ g(x) = x + 3$

$D.\quad f(x) = 3x + 3,\ g(x) = x^2 + 4$

5. 10 pts

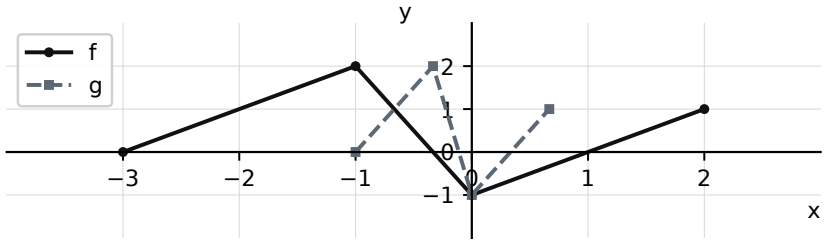
6. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x - 3)$

$B.\quad g(x) = f(3x)$

$C.\quad g(x) = f(x/3)$

$D.\quad g(x) = 3f(x)$



6. 10 pts

Test continues on the next sheet.

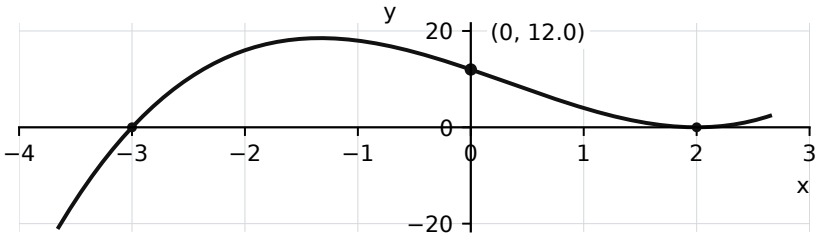
MATH 101 **79298**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
8. Solve exactly. $8^{x-2} = 4^{x+2}$	8. 10 pts
9. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	9. 10 pts
10. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	10. 10 pts

79298

TEST PAGE 1

1. (a)
$(2, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 2$ 1 pts

(d)
$(\frac{9}{4}, 0)$ 1 pts

(e)
$x \to 2^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

2. (a)
$[-16, \infty)$ 2 pts

(b)
$[-2, \infty)$ 2 pts

(c)
$(-12, 0)$ 1 pts

(d)
$(0, 2)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -16^+ : f(x) \to -2$ 1 pts

3. (a)
4 5 pts

(b)
$\frac{3}{2}$ 5 pts

TEST PAGE 2

4. (a)
C
3 pts

(b)
A
2 pts

(c)
D
3 pts

(d)
B
2 pts

5.
A
10 pts

6.
B
10 pts

TEST PAGE 3

7. (a)
$\mathbb{R} \setminus \{-2, 5\}$ 2 pts

(b)
$(-2, -\frac{1}{7})$ 2 pts

(c)
$x = 5$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-3, 0)$ 2 pts

(f)
$(0, -\frac{3}{5})$ 2 pts

8.
$x = 10$
10 pts

9.
$1(x + 3)(x - 2)^2$
10 pts

10.
$x = -1, -1 + \sqrt{3}, -\sqrt{3} - 1$
10 pts

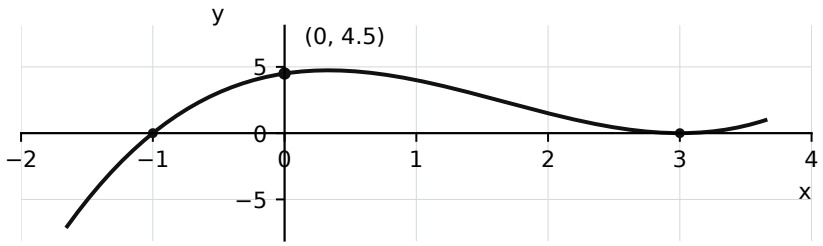
MATH 101 **85509**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.  	1. 10 pts
2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+4} - 6$	2. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
3. Which pair satisfies h(x) = f(g(x))? $h(x) = (4x+4)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$ $B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$	3. 10 pts

Test continues on the other side of the sheet.

MATH 101 **85509**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-3) + 2$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	6. 10 pts

Test continues on the next sheet.

MATH 101 **85509**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

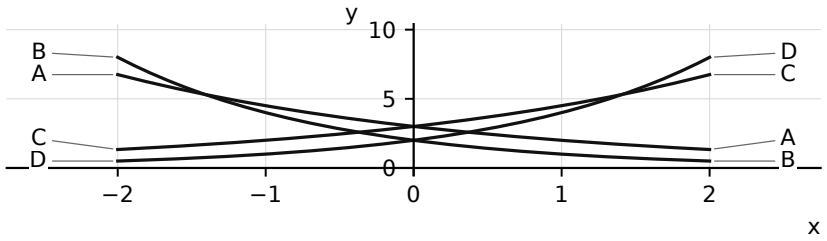
7. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $3(\frac{3}{2})^x$

(*b*) $3(\frac{2}{3})^x$

(*c*) $2(2)^x$

(*d*) $2(\frac{1}{2})^x$



7. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

8. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.25) + \log(400)$

(*b*) $\frac{\log_7(64)}{\log_7(16)}$

8. (a) 5 pts

(b) 5 pts

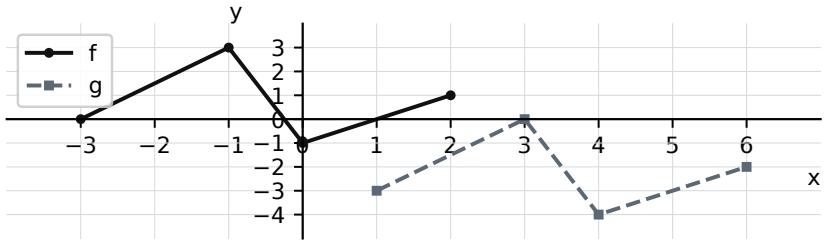
9. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x-4)+3$

B. $g(x) = f(x+4)-3$

C. $g(x) = f(x-4)-3$

D. $g(x) = f(x+4)+3$



9. 10 pts

10. Solve exactly.

$\log_2(x-5) + \log_2(2) = 2$

10. 10 pts

85509

TEST PAGE 1

1.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

2. (a)

$[-4, \infty)$ 2 pts

(b)

$[-6, \infty)$ 2 pts

(c)

$(5, 0)$ 1 pts

(d)

$(0, -2)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -4^+ : f(x) \to -6$ 1 pts

3.

A

10 pts

TEST PAGE 2

4. (a)

$\mathbb{R} \setminus \{-3, 6\}$ 2 pts

(b)

$(-3, -\frac{1}{9})$ 2 pts

(c)

$x = 6$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-4, 0)$ 2 pts

(f)

$(0, -\frac{2}{3})$ 2 pts

5. (a)

$(3, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 3$ 1 pts

(d)

$(\frac{13}{4}, 0)$ 1 pts

(e)

$x \to 3^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

6.

$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 10 pts

TEST PAGE 3

7. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

8. (a)

2 5 pts

(b)

$\frac{3}{2}$ 5 pts

9.

C

10 pts

10.

$x = 7$

10 pts

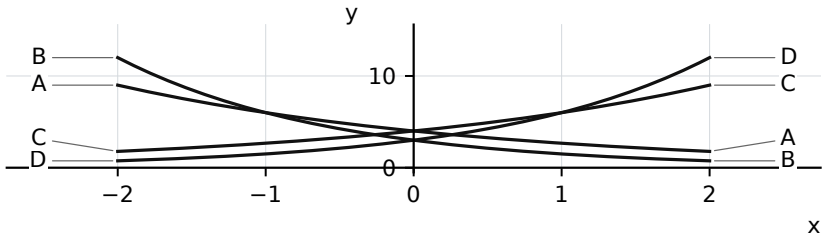
MATH 101 **58263**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	1. 10 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 4(\frac{3}{2})^x$ $(b)\quad 4(\frac{2}{3})^x$ $(c)\quad 3(2)^x$ $(d)\quad 3(\frac{1}{2})^x$ 	2. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
3. Which pair satisfies h(x) = f(g(x))? $h(x) = (4x + 4)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$ $B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$	3. 10 pts
4. Solve the nonlinear system. $y = x^2$ $y = 4x + 32$	4. 10 pts

Test continues on the other side of the sheet.

MATH 101 **58263**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

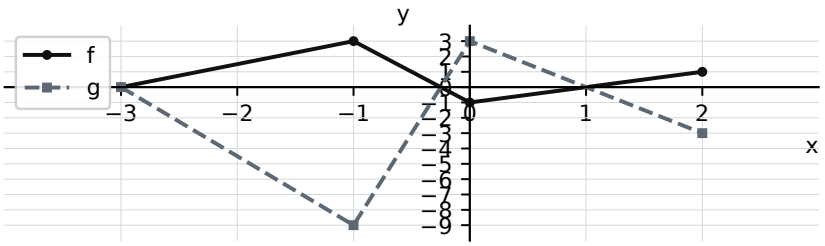
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = 3f(x)$

B. $g(x) = f(-3x)$

C. $g(x) = -3f(x)$

D. $g(x) = f(x) - 3$



5.

10 pts

6. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$$f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$$

6. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

Test continues on the next sheet.

MATH 101 **58263**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+9} - 6$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-3) + 2$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
9. Evaluate exactly. Here log means base 10. (a) $\log(0.8) + \log(1250)$ (b) $\dfrac{\log_7(64)}{\log_7(16)}$	9. (a) 5 pts (b) 5 pts

Test continues on the other side of the sheet.

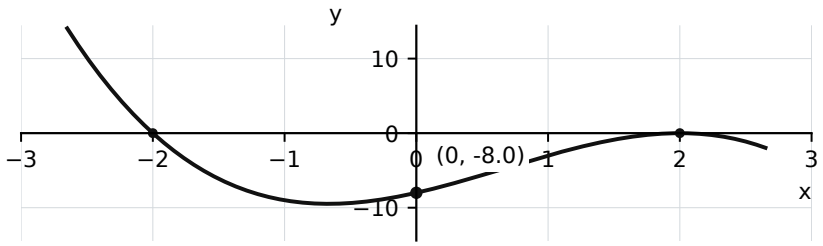
MATH 101 **58263**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	10. 10 pts

58263

TEST PAGE 1

1.

$x = -2,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

2. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

3.

A

10 pts

4.

$(-4, 16); (8, 64)$

10 pts

TEST PAGE 2

5.

C

10 pts

6. (a)

$\mathbb{R} \setminus \{-3, 6\}$ 2 pts

(b)

$(-3,\ -\frac{1}{9})$ 2 pts

(c)

$x = 6$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-4, 0)$ 2 pts

(f)

$(0,\ -\frac{2}{3})$ 2 pts

TEST PAGE 3

7. (a)

$[-9, \infty)$ 2 pts

(b)

$[-6, \infty)$ 2 pts

(c)

$(0, 0)$ 1 pts

(d)

$(0, 0)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^{+} : f(x) \to -6$ 1 pts

8. (a)

$(3, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 3$ 1 pts

(d)

$(\frac{28}{9}, 0)$ 1 pts

(e)

$x \to 3^{+} : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

9. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

58263

TEST PAGE 4

10.

$-1(x+2)(x-2)^2$

10 pts

MATH 101 **83016**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
2. Solve exactly. $\sqrt{(x-1)^2+9} = 5$	2. 10 pts
3. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-3)+2$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **83016**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Which pair satisfies h(x) = f(g(x))?

$h(x) = (4x + 4)^2 + 5$

$A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$

$B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$

$C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$

$D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$

4.

10 pts

5. Evaluate exactly. Here log means base 10.

$(a)\quad \log(0.05) + \log(200000)$

$(b)\quad \dfrac{\log_7(64)}{\log_7(16)}$

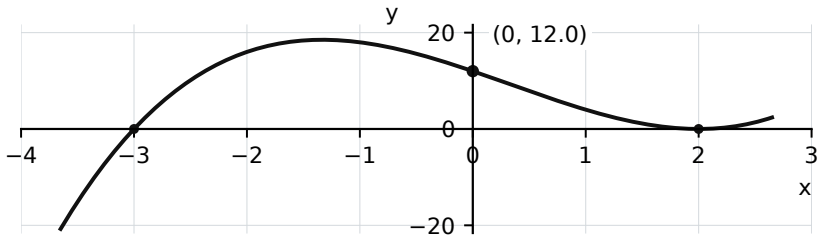
5. (a)

5 pts

(b)

5 pts

6. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



6.

10 pts

7. Find all zeros.

$p(x) = x^3 + 3x^2 - 2$

7.

10 pts

Test continues on the next sheet.

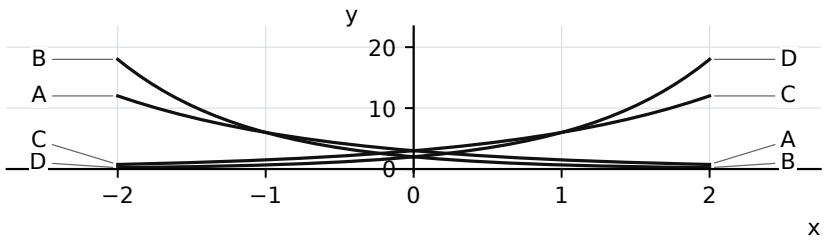
MATH 101 **83016**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 2\sqrt{x+16} - 6$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
9. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(2)^x$ $(b)\quad 3(\frac{1}{2})^x$ $(c)\quad 2(3)^x$ $(d)\quad 2(\frac{1}{3})^x$ 	9. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

Test continues on the other side of the sheet.

MATH 101 **83016**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

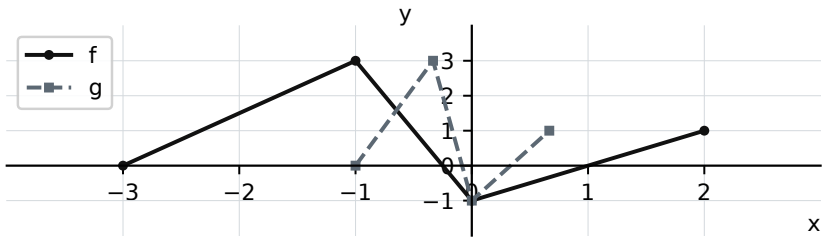
10. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x/3)$

B. $g(x) = 3f(x)$

C. $g(x) = f(3x)$

D. $g(x) = f(x - 3)$



10.

10 pts

83016

TEST PAGE 1

1. (a)
$\mathbb{R} \setminus \{-3, 6\}$ 2 pts

(b)
$(-3, -\frac{1}{9})$ 2 pts

(c)
$x = 6$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-4, 0)$ 2 pts

(f)
$(0, -\frac{2}{3})$ 2 pts

2.
$x = -3, 5$
10 pts

3. (a)
$(3, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 3$ 1 pts

(d)
$(\frac{49}{16}, 0)$ 1 pts

(e)
$x \to 3^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

TEST PAGE 2

4.
A
10 pts

5. (a)
4 5 pts

(b)
$\frac{3}{2}$ 5 pts

6.
$1(x+3)(x-2)^2$
10 pts

7.
$x = -1, -1+\sqrt{3}, -\sqrt{3}-1$
10 pts

TEST PAGE 3

8. (a)
$[-16, \infty)$ 2 pts

(b)
$[-6, \infty)$ 2 pts

(c)
$(-7, 0)$ 1 pts

(d)
$(0, 2)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -16^+ : f(x) \to -6$ 1 pts

9. (a)
C
3 pts

(b)
A
2 pts

(c)
D
3 pts

(d)
B
2 pts

83016

TEST PAGE 4

10.

C

10 pts

MATH 101 **98061**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+4} - 12$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
2. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	2. 10 pts
3. Solve exactly. $8^{x-4} = 4^{x+1}$	3. 10 pts
4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **98061**

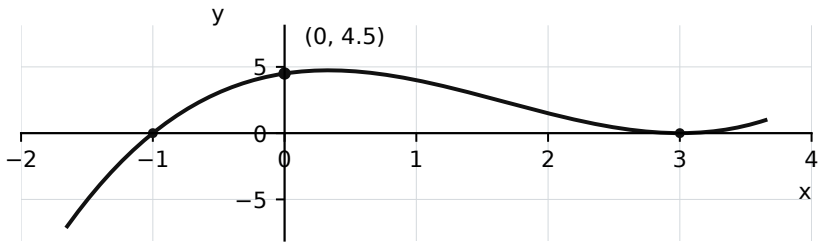
Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



5.

10 pts

6. Which pair satisfies h(x) = f(g(x))?

$h(x) = (5x + 5)^2 + 6$

$A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$

$B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$

$C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$

$D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$

6.

10 pts

7. Evaluate exactly. Here log means base 10.

$(a)\quad \log(0.25) + \log(400)$

$(b)\quad \frac{\log_8(125)}{\log_8(25)}$

7. (a)

5 pts

(b)

5 pts

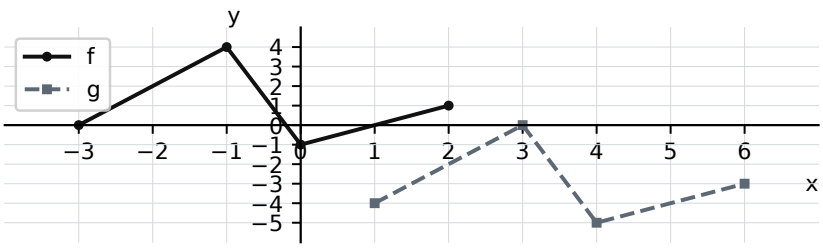
8. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x - 4) - 4$

$B.\quad g(x) = f(x + 4) + 4$

$C.\quad g(x) = f(x - 4) + 4$

$D.\quad g(x) = f(x + 4) - 4$



8.

10 pts

Test continues on the next sheet.

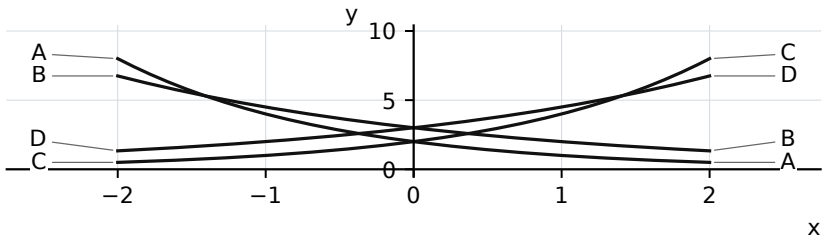
MATH 101 **98061**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-4)+3$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
10. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(2)^x$ $(b)\quad 2(\frac{1}{2})^x$ $(c)\quad 3(\frac{3}{2})^x$ $(d)\quad 3(\frac{2}{3})^x$ 	10. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

98061

TEST PAGE 1

1. (a)
$[-4, \infty)$ 2 pts

(b)
$[-12, \infty)$ 2 pts

(c)
$(12, 0)$ 1 pts

(d)
$(0, -6)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -4^+ : f(x) \to -12$ 1 pts

2.
$x = 3, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2} - \frac{1}{2}$ 10 pts

3.
$x = 14$ 10 pts

4. (a)
$\mathbb{R} \setminus \{-4, 7\}$ 2 pts

(b)
$(-4, -\frac{1}{11})$ 2 pts

(c)
$x = 7$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-5, 0)$ 2 pts

(f)
$(0, -\frac{5}{7})$ 2 pts

TEST PAGE 2

5.
$\frac{1}{2}(x+1)(x-3)^2$

10 pts

6.
D

10 pts

7. (a)
2 5 pts

(b)
$\frac{3}{2}$ 5 pts

8.
A

10 pts

TEST PAGE 3

9. (a)
$(4, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 4$ 1 pts

(d)
$(\frac{33}{8}, 0)$ 1 pts

(e)
$x \to 4^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

10. (a)
C

3 pts

(b)
A

2 pts

(c)
D

3 pts

(d)
B

2 pts

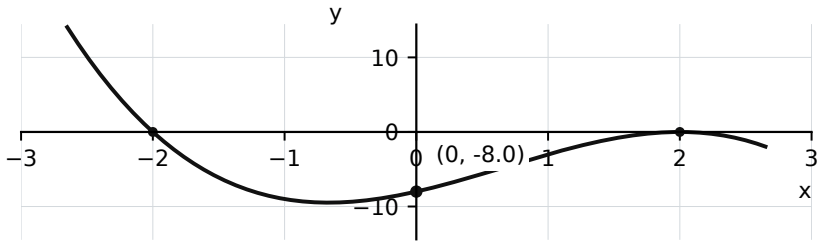
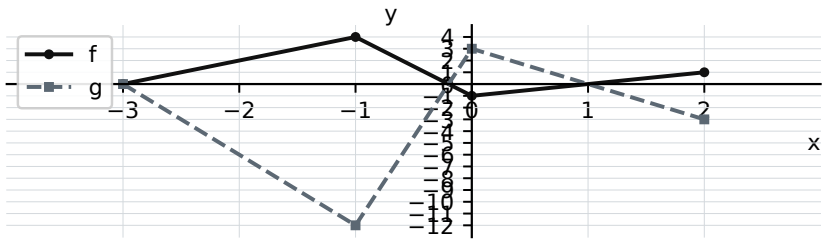
MATH 101 **70802**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve exactly. $\log_2(x-4) - \log_2(x-8) = 1$	1. 10 pts
2. Evaluate exactly. Here log means base 10. (a) $\log(0.8) + \log(1250)$ (b) $\frac{\log_8(125)}{\log_8(25)}$	2. (a) 5 pts (b) 5 pts
3. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	3. 10 pts
4. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.$ $g(x) = -3f(x)$ $B.$ $g(x) = f(x) - 3$ $C.$ $g(x) = 3f(x)$ $D.$ $g(x) = f(-3x)$ 	4. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+9} - 12$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
7. Which pair satisfies h(x) = f(g(x))? $h(x) = (5x+5)^2 + 6$ $A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$ $B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$ $C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$ $D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$	7. 10 pts

Test continues on the next sheet.

MATH 101 **70802**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

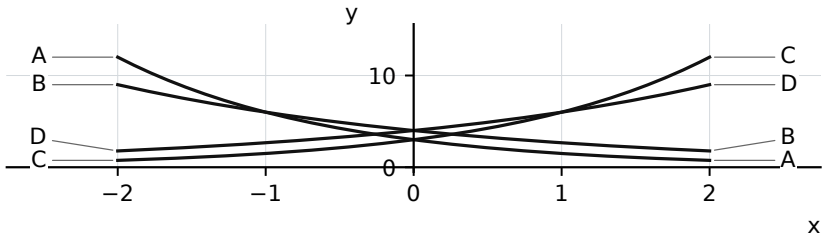
8. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $3(2)^x$

(b) $3(\frac{1}{2})^x$

(c) $4(\frac{3}{2})^x$

(d) $4(\frac{2}{3})^x$



8. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

9. Find all zeros.

$p(x) = x^3 + 5x^2 + 7x + 2$

9. 10 pts

10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_3(x - 4) + 3$

10. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 2 pts

(f) 2 pts

70802

TEST PAGE 1

1.

$x = 12$

10 pts

2. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

3.

$-1(x+2)(x-2)^2$

10 pts

4.

A

10 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{-4, 7\}$ 2 pts

(b)

$(-4, -\frac{1}{11})$ 2 pts

(c)

$x = 7$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-5, 0)$ 2 pts

(f)

$(0, -\frac{5}{7})$ 2 pts

6. (a)

$[-9, \infty)$ 2 pts

(b)

$[-12, \infty)$ 2 pts

(c)

$(7, 0)$ 1 pts

(d)

$(0, -3)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -12$ 1 pts

7.

D

10 pts

TEST PAGE 3

8. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

9.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

10. (a)

$(4, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 4$ 1 pts

(d)

$(\frac{109}{27}, 0)$ 1 pts

(e)

$x \to 4^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

MATH 101 **46900**

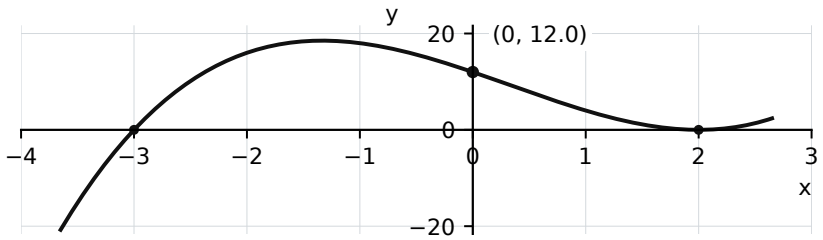
Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



1.

10 pts

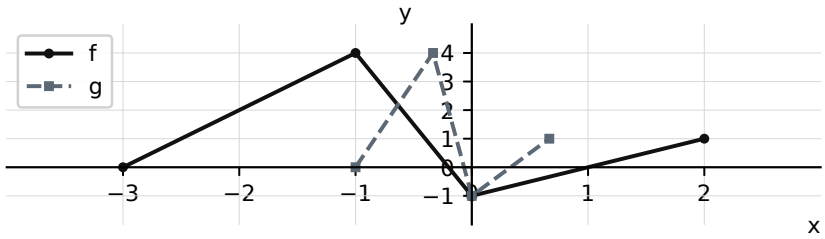
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(3x)$

B. $g(x) = f(x - 3)$

C. $g(x) = f(x/3)$

D. $g(x) = 3f(x)$



2.

10 pts

3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$$f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$$

3. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

Test continues on the other side of the sheet.

MATH 101 **46900**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Evaluate exactly. Here log means base 10. (a) $\log(0.05) + \log(200000)$ (b) $\frac{\log_8(125)}{\log_8(25)}$	4. (a) 5 pts (b) 5 pts
5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 3\sqrt{x+16} - 12$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
6. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-4) + 3$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
7. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	7. 10 pts

Test continues on the next sheet.

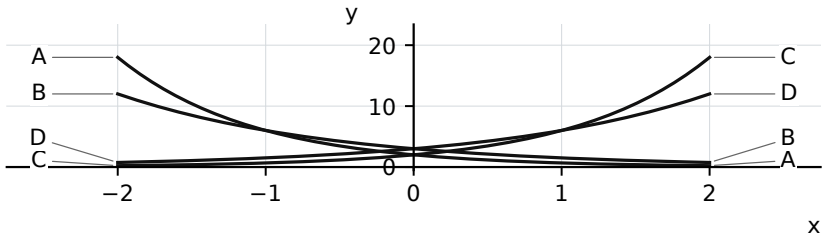
MATH 101 **46900**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(3)^x$ $(b)\quad 2(\frac{1}{3})^x$ $(c)\quad 3(2)^x$ $(d)\quad 3(\frac{1}{2})^x$  	8. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
9. Solve the nonlinear system. $x^2 + y^2 = 5$ $y = x + 1$	9. 10 pts
10. Which pair satisfies h(x) = f(g(x))? $h(x) = (5x + 5)^2 + 6$ $A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$ $B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$ $C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$ $D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$	10. 10 pts

46900

TEST PAGE 1

1.

$1(x+3)(x-2)^2$

10 pts

2.

A

10 pts

3. (a)

$\mathbb{R} \setminus \{-4, 7\}$ 2 pts

(b)

$(-4, -\frac{1}{11})$ 2 pts

(c)

$x = 7$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-5, 0)$ 2 pts

(f)

$(0, -\frac{5}{7})$ 2 pts

TEST PAGE 2

4. (a)

4 5 pts

(b)

$\frac{3}{2}$ 5 pts

5. (a)

$[-16, \infty)$ 2 pts

(b)

$[-12, \infty)$ 2 pts

(c)

$(0, 0)$ 1 pts

(d)

$(0, 0)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -16^+ : f(x) \to -12$ 1 pts

6. (a)

$(4, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 4$ 1 pts

(d)

$(\frac{257}{64}, 0)$ 1 pts

(e)

$x \to 4^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

7.

$x = -1, -1 + \sqrt{3}, -\sqrt{3} - 1$

10 pts

TEST PAGE 3

8. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

9.

$(1, 2); (-2, -1)$

10 pts

10.

D

10 pts

MATH 101 **81643**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

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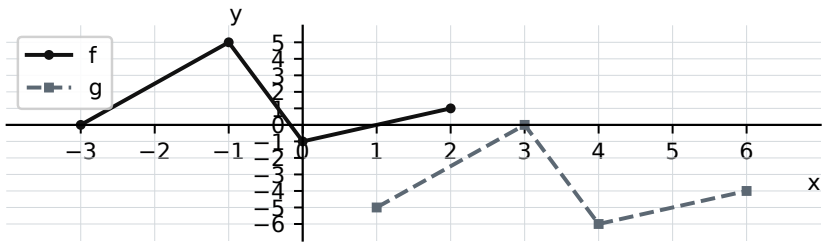
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x-4)+5$

$B.\quad g(x) = f(x+4)-5$

$C.\quad g(x) = f(x+4)+5$

$D.\quad g(x) = f(x-4)-5$



1.

10 pts

---

2. Evaluate exactly. Here log means base 10.

$(a)\quad \log(0.25)+\log(400)$

$(b)\quad \dfrac{\log_9(216)}{\log_9(36)}$

2. (a)

5 pts

(b)

5 pts

---

3. Find all zeros.

$p(x) = x^3 - 2x^2 - 4x + 3$

3.

10 pts

---

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept.

$f(x) = \dfrac{(x+5)(x+6)}{(x-8)(x+5)}$

4. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

Test continues on the other side of the sheet.

MATH 101 **81643**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-5)+1$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+4}-20$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
7. Which pair satisfies h(x) = f(g(x))? $h(x) = (6x+6)^2+7$ $A.\quad f(x)=6x^2+7,\ g(x)=x+6$ $B.\quad f(x)=x^2+7,\ g(x)=6x+6$ $C.\quad f(x)=x^2+6,\ g(x)=6x+7$ $D.\quad f(x)=6x+6,\ g(x)=x^2+7$	7. 10 pts

Test continues on the next sheet.

MATH 101 **81643**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Solve exactly.

$\sqrt{x+2} = x - 2$

8.

10 pts

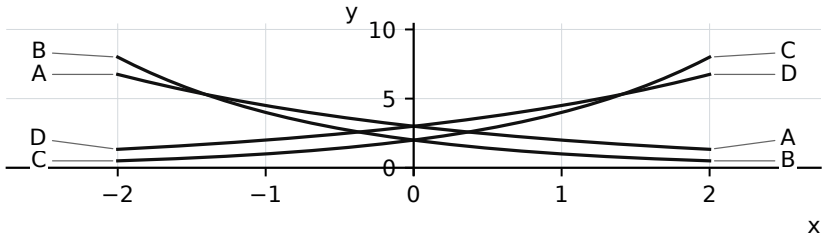
9. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(2)^x$

(b) $3(\frac{2}{3})^x$

(c) $3(\frac{3}{2})^x$

(d) $2(\frac{1}{2})^x$



9. (a)

3 pts

(b)

2 pts

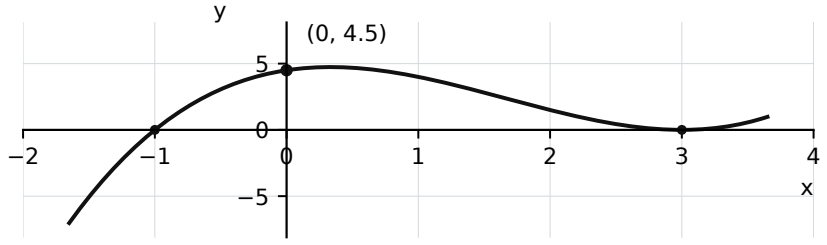
(c)

3 pts

(d)

2 pts

10. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



10.

10 pts

81643

TEST PAGE 1

1.

D

10 pts

2. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

3.

$x = 3,\ -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$

10 pts

4. (a)

$\mathbb{R} \setminus \{-5, 8\}$

2 pts

(b)

$(-5,\ -\frac{1}{13})$

2 pts

(c)

$x = 8$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-6, 0)$

2 pts

(f)

$(0,\ -\frac{3}{4})$

2 pts

TEST PAGE 2

5. (a)

$(5, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 5$

1 pts

(d)

$(\frac{11}{2}, 0)$

1 pts

(e)

$x \to 5^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

6. (a)

$[-4, \infty)$

2 pts

(b)

$[-20, \infty)$

2 pts

(c)

$(21, 0)$

1 pts

(d)

$(0, -12)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -4^+ : f(x) \to -20$

1 pts

7.

B

10 pts

TEST PAGE 3

8.

$x = \frac{\sqrt{17}}{2} + \frac{5}{2}$

10 pts

9. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

10.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

MATH 101 **99545**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.8)+\log(1250)$ $(b)\quad \frac{\log_9(216)}{\log_9(36)}$	1. (a) 5 pts (b) 5 pts
2. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x)=\frac{(x+5)(x+6)}{(x-8)(x+5)}$	2. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x)=4\sqrt{x+9}-20$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

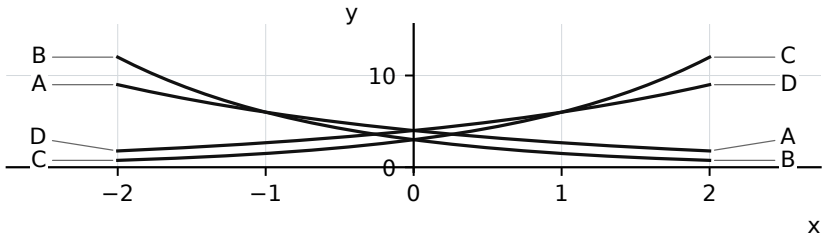
4. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $3(2)^x$

(b) $4(\frac{2}{3})^x$

(c) $4(\frac{3}{2})^x$

(d) $3(\frac{1}{2})^x$



4. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_3(x-5)+1$

5. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 2 pts

(f) 2 pts

Test continues on the next sheet.

MATH 101 **99545**

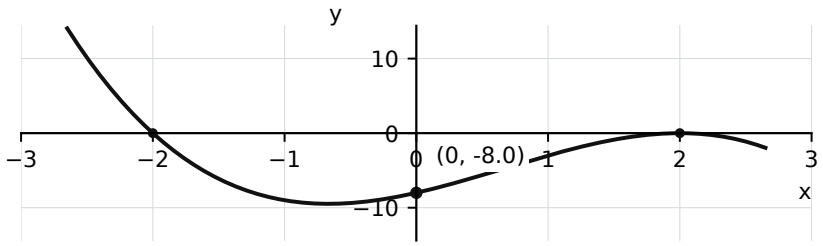
Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

6. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



6.

10 pts

7. Find all zeros.

$p(x) = x^3 + 5x^2 + 7x + 2$

7.

10 pts

8. Solve exactly.

$8^{x-1} = 4^{x+4}$

8.

10 pts

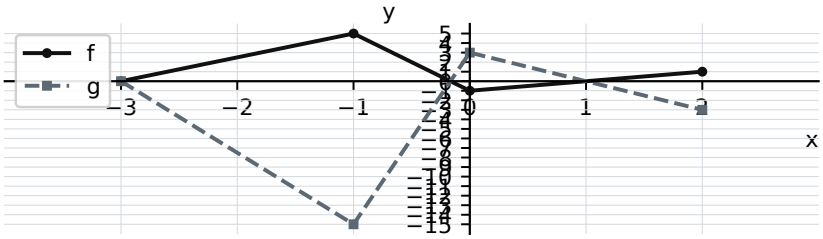
9. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = 3f(x)$

$B.\quad g(x) = f(-3x)$

$C.\quad g(x) = f(x) - 3$

$D.\quad g(x) = -3f(x)$



9.

10 pts

10. Which pair satisfies h(x) = f(g(x))?

$h(x) = (6x + 6)^2 + 7$

$A.\quad f(x) = 6x^2 + 7,\ g(x) = x + 6$

$B.\quad f(x) = x^2 + 7,\ g(x) = 6x + 6$

$C.\quad f(x) = x^2 + 6,\ g(x) = 6x + 7$

$D.\quad f(x) = 6x + 6,\ g(x) = x^2 + 7$

10.

10 pts

99545

TEST PAGE 1

1. (a)

3 — 5 pts

(b)

$\frac{3}{2}$ — 5 pts

2. (a)

$\mathbb{R} \setminus \{-5, 8\}$ — 2 pts

(b)

$(-5, -\frac{1}{13})$ — 2 pts

(c)

$x = 8$ — 1 pts

(d)

$y = 1$ — 1 pts

(e)

$(-6, 0)$ — 2 pts

(f)

$(0, -\frac{3}{4})$ — 2 pts

3. (a)

$[-9, \infty)$ — 2 pts

(b)

$[-20, \infty)$ — 2 pts

(c)

$(16, 0)$ — 1 pts

(d)

$(0, -8)$ — 1 pts

(e)

$x \to \infty : f(x) \to \infty$ — 3 pts

(f)

$x \to -9^+ : f(x) \to -20$ — 1 pts

TEST PAGE 2

4. (a)

C — 3 pts

(b)

A — 2 pts

(c)

D — 3 pts

(d)

B — 2 pts

5. (a)

$(5, \infty)$ — 2 pts

(b)

$(-\infty, \infty)$ — 2 pts

(c)

$x = 5$ — 1 pts

(d)

$(\frac{16}{3}, 0)$ — 1 pts

(e)

$x \to 5^+ : f(x) \to -\infty$ — 2 pts

(f)

$x \to \infty : f(x) \to \infty$ — 2 pts

TEST PAGE 3

6.

$-1(x+2)(x-2)^2$

10 pts

7.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ — 10 pts

8.

$x = 11$

10 pts

9.

D

10 pts

10.

B

10 pts

MATH 101 **45589**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 4\sqrt{x+16} - 20$	1. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
2. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.05) + \log(200000)$ $(b)\quad \frac{\log_9(216)}{\log_9(36)}$	2. (a) 5 pts (b) 5 pts
3. Solve exactly. $\log_3(x-1) + \log_3(x-9) = 2$	3. 10 pts
4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+5)(x+6)}{(x-8)(x+5)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

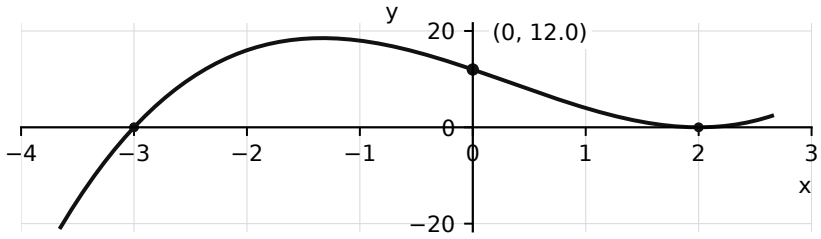
MATH 101 **45589**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x - 5) + 1$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	6. 10 pts
7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	7. 10 pts

Test continues on the next sheet.

MATH 101 **45589**

Diagnostic Exam

Name: _____ Date: _____

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Write final answers in the space provided in the right column.

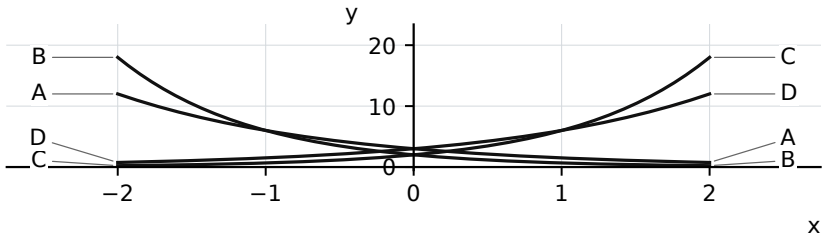
8. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(3)^x$

(b) $3(\frac{1}{2})^x$

(c) $3(2)^x$

(d) $2(\frac{1}{3})^x$



8. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

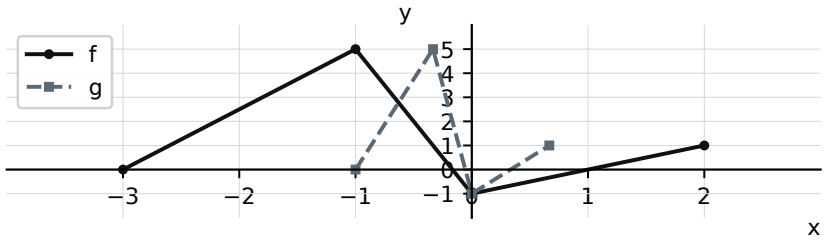
9. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.$ $g(x) = f(x/3)$

$B.$ $g(x) = 3f(x)$

$C.$ $g(x) = f(x-3)$

$D.$ $g(x) = f(3x)$



9.

10 pts

10. Which pair satisfies h(x) = f(g(x))?

$h(x) = (6x+6)^2 + 7$

$A.$ $f(x) = 6x^2 + 7,\ g(x) = x + 6$

$B.$ $f(x) = x^2 + 7,\ g(x) = 6x + 6$

$C.$ $f(x) = x^2 + 6,\ g(x) = 6x + 7$

$D.$ $f(x) = 6x + 6,\ g(x) = x^2 + 7$

10.

10 pts

45589

TEST PAGE 1

1. (a)
$[-16, \infty)$ 2 pts

(b)
$[-20, \infty)$ 2 pts

(c)
$(9, 0)$ 1 pts

(d)
$(0, -4)$ 1 pts

(e)
$x \to \infty : f(x) \to \infty$ 3 pts

(f)
$x \to -16^+ : f(x) \to -20$ 1 pts

2. (a)
4 5 pts

(b)
$\frac{3}{2}$ 5 pts

3.
$x = 10$
10 pts

4. (a)
$\mathbb{R} \setminus \{-5, 8\}$ 2 pts

(b)
$(-5, -\frac{1}{13})$ 2 pts

(c)
$x = 8$ 1 pts

(d)
$y = 1$ 1 pts

(e)
$(-6, 0)$ 2 pts

(f)
$(0, -\frac{3}{4})$ 2 pts

TEST PAGE 2

5. (a)
$(5, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 5$ 1 pts

(d)
$(\frac{21}{4}, 0)$ 1 pts

(e)
$x \to 5^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

6.
$x = -1, -1 + \sqrt{3}, -\sqrt{3} - 1$
10 pts

7.
$1(x + 3)(x - 2)^2$
10 pts

TEST PAGE 3

8. (a)
C
3 pts

(b)
A
2 pts

(c)
D
3 pts

(d)
B
2 pts

9.
D
10 pts

10.
B
10 pts

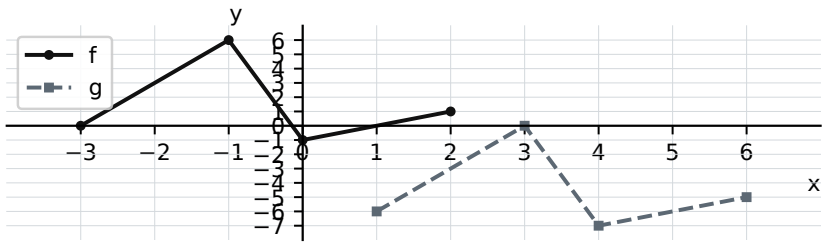
MATH 101 **25484**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x-4) - 6$ $B.\quad g(x) = f(x+4) + 6$ $C.\quad g(x) = f(x-4) + 6$ $D.\quad g(x) = f(x+4) - 6$ 	1. 10 pts
2. Solve the nonlinear system. $y = x^2$ $y = 3x + 18$	2. 10 pts
3. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.25) + \log(400)$ $(b)\quad \frac{\log_{10}(343)}{\log_{10}(49)}$	3. (a) 5 pts (b) 5 pts
4. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	4. 10 pts

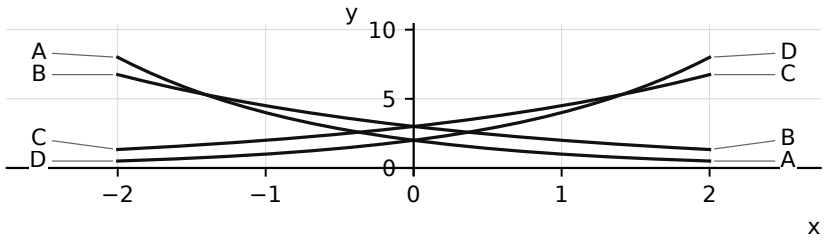
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Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x+4} - 30$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
6. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(\frac{3}{2})^x$ $(b)\quad 2(\frac{1}{2})^x$ $(c)\quad 2(2)^x$ $(d)\quad 3(\frac{2}{3})^x$ 	6. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

Test continues on the next sheet.

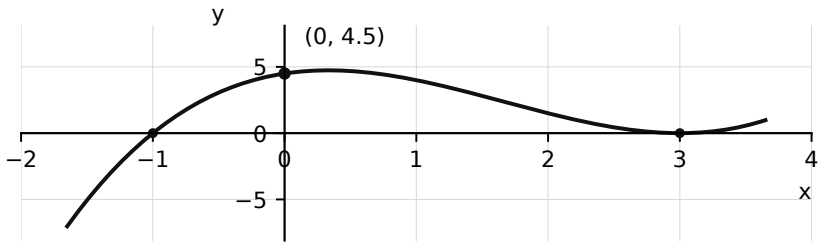
MATH 101 **25484**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	7. 10 pts
8. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+6)(x+7)}{(x-9)(x+6)}$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
9. Which pair satisfies h(x) = f(g(x))? $h(x) = (7x+7)^2 + 8$ $A.\quad f(x) = 7x+7,\ g(x) = x^2+8$ $B.\quad f(x) = 7x^2+8,\ g(x) = x+7$ $C.\quad f(x) = x^2+8,\ g(x) = 7x+7$ $D.\quad f(x) = x^2+7,\ g(x) = 7x+8$	9. 10 pts

Test continues on the other side of the sheet.

MATH 101 **25484**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 6) + 2$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

25484

TEST PAGE 1

1.

A

10 pts

2.

$(-3, 9); (6, 36)$

10 pts

3. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

4.

$x = 3,\ -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$

10 pts

TEST PAGE 2

5. (a)

$[-4, \infty)$

2 pts

(b)

$[-30, \infty)$

2 pts

(c)

$(32, 0)$

1 pts

(d)

$(0, -20)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -4^+ : f(x) \to -30$

1 pts

6. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

TEST PAGE 3

7.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

8. (a)

$\mathbb{R} \setminus \{-6, 9\}$

2 pts

(b)

$(-6, -\frac{1}{15})$

2 pts

(c)

$x = 9$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-7, 0)$

2 pts

(f)

$(0, -\frac{7}{9})$

2 pts

9.

C

10 pts

25484

TEST PAGE 4

10. (a)
$(6, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 6$ 1 pts

(d)
$(\frac{25}{4}, 0)$ 1 pts

(e)
$x \to 6^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

MATH 101 **32681**

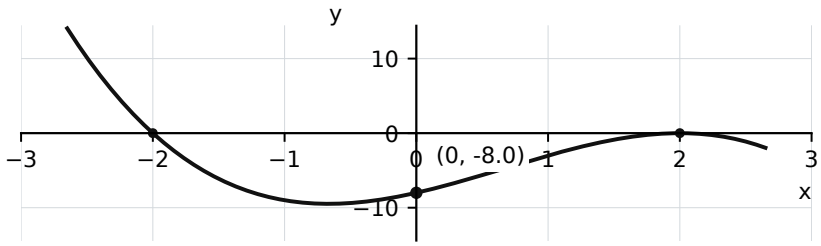
Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



1.

10 pts

2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 5\sqrt{x+9} - 30$

2. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

3 pts

(f)

1 pts

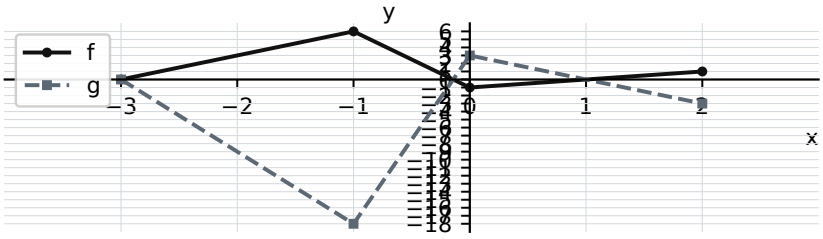
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = -3f(x)$

B. $g(x) = f(x) - 3$

C. $g(x) = 3f(x)$

D. $g(x) = f(-3x)$



3.

10 pts

Test continues on the other side of the sheet.

MATH 101 **32681**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.8)+\log(1250)$ $(b)\quad \frac{\log_{10}(343)}{\log_{10}(49)}$	4. (a) 5 pts (b) 5 pts
5. Solve exactly. $\sqrt{(x-2)^2+35}=6$	5. 10 pts
6. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x)=\frac{(x+6)(x+7)}{(x-9)(x+6)}$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
7. Which pair satisfies h(x) = f(g(x))? $h(x)=(7x+7)^2+8$ $A.\quad f(x)=7x+7,\ g(x)=x^2+8$ $B.\quad f(x)=7x^2+8,\ g(x)=x+7$ $C.\quad f(x)=x^2+8,\ g(x)=7x+7$ $D.\quad f(x)=x^2+7,\ g(x)=7x+8$	7. 10 pts

Test continues on the next sheet.

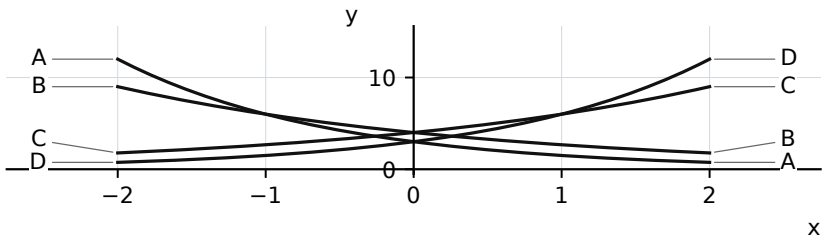
MATH 101 **32681**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-6)+2$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
9. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 4(\frac{3}{2})^x$ $(b)\quad 3(\frac{1}{2})^x$ $(c)\quad 3(2)^x$ $(d)\quad 4(\frac{2}{3})^x$ 	9. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
10. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	10. 10 pts

32681

TEST PAGE 1

1.

$-1(x+2)(x-2)^2$

10 pts

2. (a)

$[-9, \infty)$ 2 pts

(b)

$[-30, \infty)$ 2 pts

(c)

$(27, 0)$ 1 pts

(d)

$(0, -15)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -30$ 1 pts

3.

A

10 pts

TEST PAGE 2

4. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

5.

$x = 1, 3$

10 pts

6. (a)

$\mathbb{R} \setminus \{-6, 9\}$ 2 pts

(b)

$(-6, -\frac{1}{15})$ 2 pts

(c)

$x = 9$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-7, 0)$ 2 pts

(f)

$(0, -\frac{7}{9})$ 2 pts

7.

C

10 pts

TEST PAGE 3

8. (a)

$(6, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 6$ 1 pts

(d)

$(\frac{55}{9}, 0)$ 1 pts

(e)

$x \to 6^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

9. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

10.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

MATH 101 **57625**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Which pair satisfies h(x) = f(g(x))? $h(x) = (7x + 7)^2 + 8$ $A.\quad f(x) = 7x + 7,\ g(x) = x^2 + 8$ $B.\quad f(x) = 7x^2 + 8,\ g(x) = x + 7$ $C.\quad f(x) = x^2 + 8,\ g(x) = 7x + 7$ $D.\quad f(x) = x^2 + 7,\ g(x) = 7x + 8$	1. 10 pts
2. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 5\sqrt{x + 16} - 30$	2. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
3. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	3. 10 pts

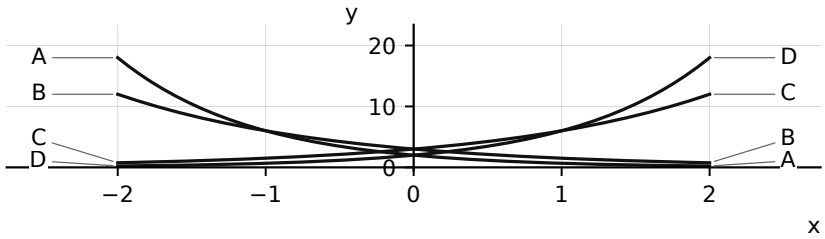
Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+6)(x+7)}{(x-9)(x+6)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(2)^x$ $(b)\quad 2(\frac{1}{3})^x$ $(c)\quad 2(3)^x$ $(d)\quad 3(\frac{1}{2})^x$ 	5. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

Test continues on the next sheet.

MATH 101 **57625**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

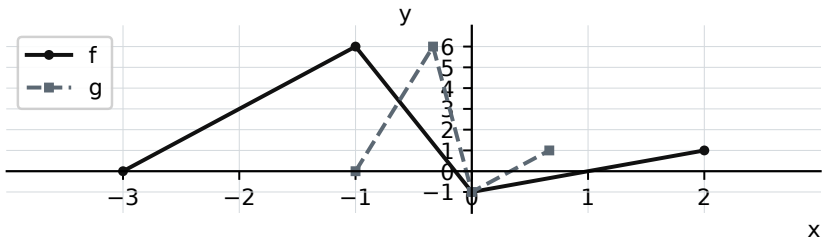
6. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(3x)$

B. $g(x) = f(x - 3)$

C. $g(x) = f(x/3)$

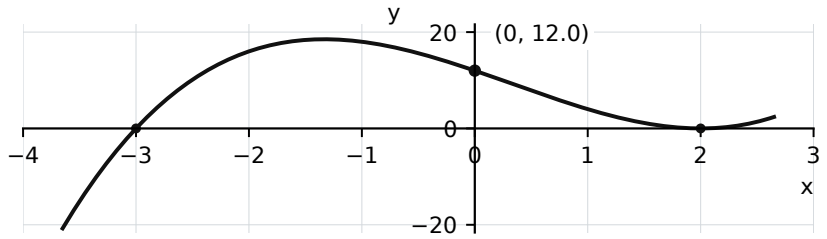
D. $g(x) = 3f(x)$



6.

10 pts

7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



7.

10 pts

8. Solve exactly.

$8^{x-4} = 4^{x+3}$

8.

10 pts

Test continues on the other side of the sheet.

MATH 101 **57625**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

9. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x - 6) + 2$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
10. Evaluate exactly. Here log means base 10. (a) $\log(0.05) + \log(200000)$ (b) $\frac{\log_{10}(343)}{\log_{10}(49)}$	10. (a) 5 pts (b) 5 pts

57625

TEST PAGE 1

1.

C

10 pts

2. (a)

$[-16, \infty)$ 2 pts

(b)

$[-30, \infty)$ 2 pts

(c)

$(20, 0)$ 1 pts

(d)

$(0, -10)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -16^+ : f(x) \to -30$ 1 pts

3.

$x = -1, -1 + \sqrt{3}, -\sqrt{3} - 1$

10 pts

TEST PAGE 2

4. (a)

$\mathbb{R} \setminus \{-6, 9\}$ 2 pts

(b)

$(-6, -\frac{1}{15})$ 2 pts

(c)

$x = 9$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-7, 0)$ 2 pts

(f)

$(0, -\frac{7}{9})$ 2 pts

5. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

TEST PAGE 3

6.

A

10 pts

7.

$1(x + 3)(x - 2)^2$

10 pts

8.

$x = 18$

10 pts

57625

TEST PAGE 4

9. (a)
$(6, \infty)$ 2 pts

(b)
$(-\infty, \infty)$ 2 pts

(c)
$x = 6$ 1 pts

(d)
$(\frac{97}{16}, 0)$ 1 pts

(e)
$x \to 6^+ : f(x) \to -\infty$ 2 pts

(f)
$x \to \infty : f(x) \to \infty$ 2 pts

10. (a)
4 5 pts

(b)
$\frac{3}{2}$ 5 pts

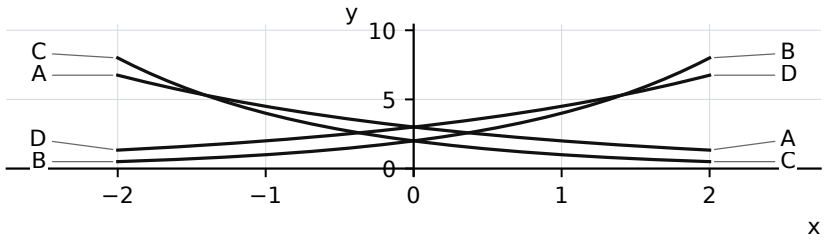
MATH 101 **65796**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve exactly. $\log_4(x-3)+\log_4(5)=2$	1. 10 pts
2. Find all zeros. $p(x)=x^3-2x^2-4x+3$	2. 10 pts
3. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(\frac{1}{2})^x$ $(b)\quad 3(\frac{2}{3})^x$ $(c)\quad 3(\frac{3}{2})^x$ $(d)\quad 2(2)^x$ 	3. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x)=\frac{(x+7)(x+8)}{(x-10)(x+7)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **65796**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

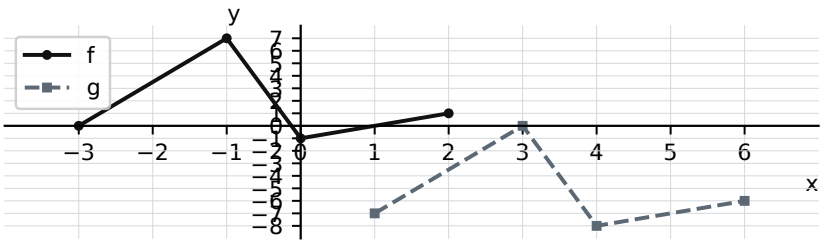
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x+4) + 7$

$B.\quad g(x) = f(x-4) + 7$

$C.\quad g(x) = f(x+4) - 7$

$D.\quad g(x) = f(x-4) - 7$



5.

10 pts

6. Which pair satisfies h(x) = f(g(x))?

$h(x) = (8x+8)^2 + 9$

$A.\quad f(x) = 8x + 8,\ g(x) = x^2 + 9$

$B.\quad f(x) = 8x^2 + 9,\ g(x) = x + 8$

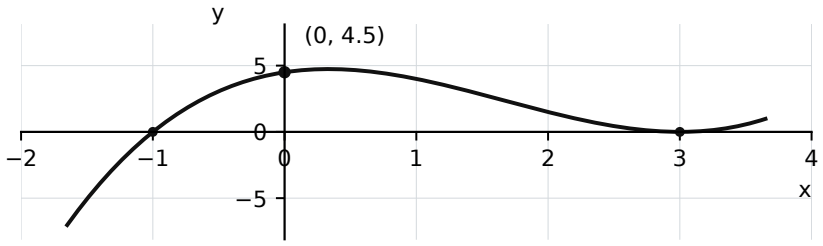
$C.\quad f(x) = x^2 + 9,\ g(x) = 8x + 8$

$D.\quad f(x) = x^2 + 8,\ g(x) = 8x + 9$

6.

10 pts

7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



7.

10 pts

Test continues on the next sheet.

MATH 101 **65796**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 7) + 3$	8. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
9. Evaluate exactly. Here log means base 10. (a) $\log(0.25) + \log(400)$ (b) $\frac{\log_{11}(512)}{\log_{11}(64)}$	9. (a) 5 pts (b) 5 pts
10. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x + 4} - 42$	10. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

65796

TEST PAGE 1

1.

$x = \frac{31}{5}$

10 pts

2.

$x = 3,\ -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$

10 pts

3. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

4. (a)

$\mathbb{R} \setminus \{-7, 10\}$

2 pts

(b)

$(-7,\ -\frac{1}{17})$

2 pts

(c)

$x = 10$

1 pts

(d)

$y = 1$

1 pts

(e)

$(-8, 0)$

2 pts

(f)

$(0,\ -\frac{4}{5})$

2 pts

TEST PAGE 2

5.

D

10 pts

6.

C

10 pts

7.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

TEST PAGE 3

8. (a)

$(7, \infty)$

2 pts

(b)

$(-\infty, \infty)$

2 pts

(c)

$x = 7$

1 pts

(d)

$(\frac{57}{8}, 0)$

1 pts

(e)

$x \to 7^+ : f(x) \to -\infty$

2 pts

(f)

$x \to \infty : f(x) \to \infty$

2 pts

9. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

10. (a)

$[-4, \infty)$

2 pts

(b)

$[-42, \infty)$

2 pts

(c)

$(45, 0)$

1 pts

(d)

$(0,\ -30)$

1 pts

(e)

$x \to \infty : f(x) \to \infty$

3 pts

(f)

$x \to -4^+ : f(x) \to -42$

1 pts

MATH 101 **14072**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

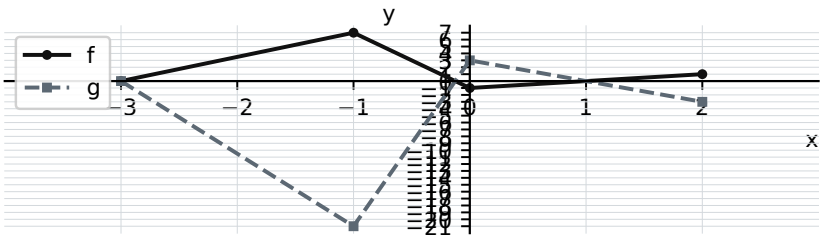
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x) - 3$

B. $g(x) = 3f(x)$

C. $g(x) = f(-3x)$

D. $g(x) = -3f(x)$



1.

10 pts

2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_3(x - 7) + 3$

2. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

Test continues on the other side of the sheet.

MATH 101 **14072**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

3. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+9} - 42$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
4. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+7)(x+8)}{(x-10)(x+7)}$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Find all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	5. 10 pts

Test continues on the next sheet.

MATH 101 **14072**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

6. Which pair satisfies h(x) = f(g(x))?

$h(x) = (8x + 8)^2 + 9$

$A.\quad f(x) = 8x + 8,\ g(x) = x^2 + 9$

$B.\quad f(x) = 8x^2 + 9,\ g(x) = x + 8$

$C.\quad f(x) = x^2 + 9,\ g(x) = 8x + 8$

$D.\quad f(x) = x^2 + 8,\ g(x) = 8x + 9$

6.

10 pts

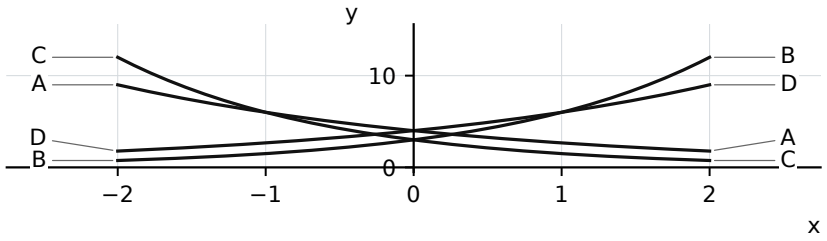
7. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(\frac{1}{2})^x$

$(b)\quad 4(\frac{2}{3})^x$

$(c)\quad 4(\frac{3}{2})^x$

$(d)\quad 3(2)^x$



7. (a)

3 pts

(b)

2 pts

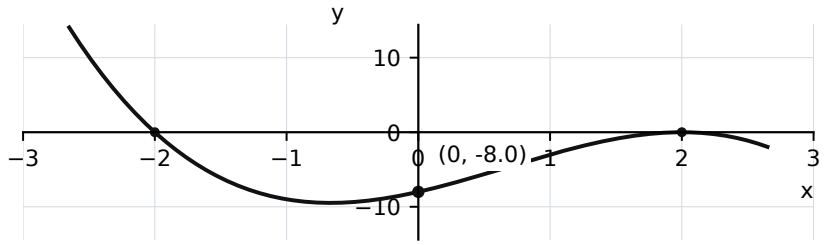
(c)

3 pts

(d)

2 pts

8. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



8.

10 pts

9. Solve the nonlinear system.

$x^2 + y^2 = 10$

$y = x + 2$

9.

10 pts

Test continues on the other side of the sheet.

MATH 101 **14072**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Evaluate exactly. Here log means base 10. (*a*) $\log(0.8) + \log(1250)$ (*b*) $\frac{\log_{11}(512)}{\log_{11}(64)}$	10. (a) 5 pts (b) 5 pts

14072

TEST PAGE 1

1.

D

10 pts

2. (a)

$(7, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 7$ 1 pts

(d)

$(\frac{190}{27}, 0)$ 1 pts

(e)

$x \to 7^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

TEST PAGE 2

3. (a)

$[-9, \infty)$ 2 pts

(b)

$[-42, \infty)$ 2 pts

(c)

$(40, 0)$ 1 pts

(d)

$(0, -24)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -42$ 1 pts

4. (a)

$\mathbb{R} \setminus \{-7, 10\}$ 2 pts

(b)

$(-7, -\frac{1}{17})$ 2 pts

(c)

$x = 10$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-8, 0)$ 2 pts

(f)

$(0, -\frac{4}{5})$ 2 pts

5.

$x = -2, -\frac{3}{2} - \frac{\sqrt{5}}{2}, -\frac{3}{2} + \frac{\sqrt{5}}{2}$ 10 pts

TEST PAGE 3

6.

C

10 pts

7. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

8.

$-1(x+2)(x-2)^2$

10 pts

9.

$(1, 3); (-3, -1)$

10 pts

14072

TEST PAGE 4

10. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

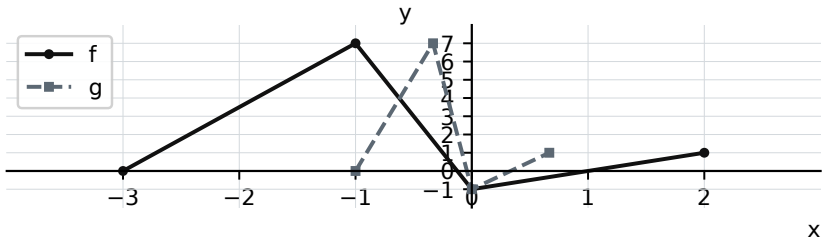
MATH 101 **15319**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate exactly. Here log means base 10. (a) $\log(0.05) + \log(200000)$ (b) $\frac{\log_{11}(512)}{\log_{11}(64)}$	1. (a) 5 pts (b) 5 pts
2. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	2. 10 pts
3. Solve exactly. $\sqrt{x+4} = x - 4$	3. 10 pts
4. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.$ $g(x) = f(x-3)$ $B.$ $g(x) = f(x/3)$ $C.$ $g(x) = 3f(x)$ $D.$ $g(x) = f(3x)$ 	4. 10 pts

Test continues on the other side of the sheet.

MATH 101 **15319**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+7)(x+8)}{(x-10)(x+7)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 6\sqrt{x+16} - 42$	6. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts
7. Which pair satisfies h(x) = f(g(x))? $h(x) = (8x+8)^2 + 9$ $A.\quad f(x) = 8x+8,\ g(x) = x^2+9$ $B.\quad f(x) = 8x^2+9,\ g(x) = x+8$ $C.\quad f(x) = x^2+9,\ g(x) = 8x+8$ $D.\quad f(x) = x^2+8,\ g(x) = 8x+9$	7. 10 pts

Test continues on the next sheet.

MATH 101 **15319**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

8. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_4(x - 7) + 3$

8. (a) 2 pts

(b) 2 pts

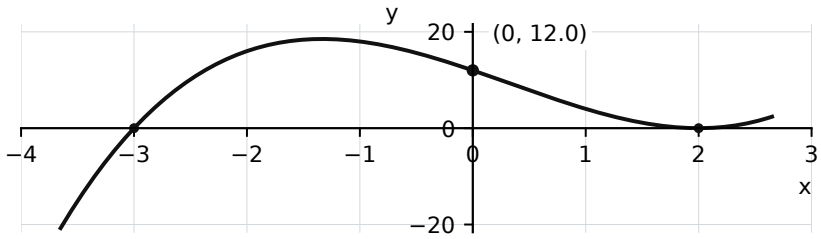
(c) 1 pts

(d) 1 pts

(e) 2 pts

(f) 2 pts

9. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



9. 10 pts

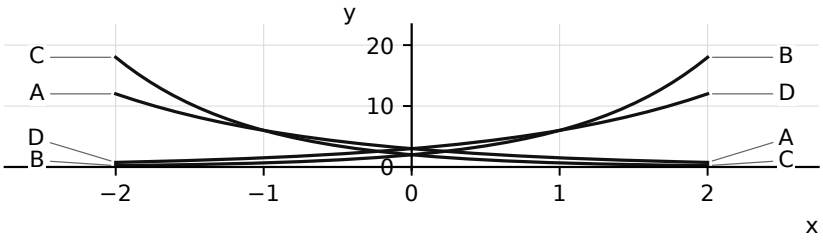
10. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 2(\frac{1}{3})^x$

$(b)\quad 3(\frac{1}{2})^x$

$(c)\quad 3(2)^x$

$(d)\quad 2(3)^x$



10. (a) 3 pts

(b) 2 pts

(c) 3 pts

(d) 2 pts

15319

TEST PAGE 1

1. (a)

4 5 pts

(b)

$\frac{3}{2}$ 5 pts

2.

$x = -1,\ -1+\sqrt{3},\ -\sqrt{3}-1$ 10 pts

3.

$x = \frac{\sqrt{33}}{2} + \frac{9}{2}$ 10 pts

4.

D

10 pts

TEST PAGE 2

5. (a)

$\mathbb{R} \setminus \{-7, 10\}$ 2 pts

(b)

$(-7,\ -\frac{1}{17})$ 2 pts

(c)

$x = 10$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-8, 0)$ 2 pts

(f)

$(0,\ -\frac{4}{5})$ 2 pts

6. (a)

$[-16, \infty)$ 2 pts

(b)

$[-42, \infty)$ 2 pts

(c)

$(33, 0)$ 1 pts

(d)

$(0,\ -18)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -16^{+} : f(x) \to -42$ 1 pts

7.

C

10 pts

TEST PAGE 3

8. (a)

$(7, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 7$ 1 pts

(d)

$(\frac{449}{64}, 0)$ 1 pts

(e)

$x \to 7^{+} : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

9.

$1(x+3)(x-2)^2$

10 pts

10. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

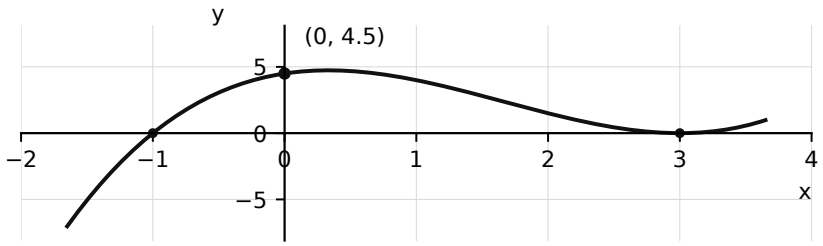
MATH 101 **36929**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.  	1. 10 pts
2. Find all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	2. 10 pts
3. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+8)(x+9)}{(x-11)(x+8)}$	3. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the other side of the sheet.

MATH 101 **36929**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

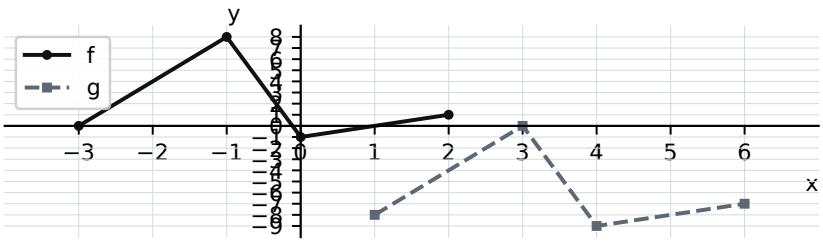
4. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x+4) - 8$

B. $g(x) = f(x-4) + 8$

C. $g(x) = f(x+4) + 8$

D. $g(x) = f(x-4) - 8$



4.

10 pts

5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_2(x-8) + 1$

5. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

2 pts

(f)

2 pts

6. Evaluate exactly. Here log means base 10.

(*a*) $\log(0.25) + \log(400)$

(*b*) $\dfrac{\log_{12}(729)}{\log_{12}(81)}$

6. (a)

5 pts

(b)

5 pts

Test continues on the next sheet.

MATH 101 **36929**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

7. Which pair satisfies h(x) = f(g(x))?

$h(x) = (9x + 9)^2 + 10$

$A.\quad f(x) = 9x^2 + 10,\ g(x) = x + 9$

$B.\quad f(x) = x^2 + 9,\ g(x) = 9x + 10$

$C.\quad f(x) = x^2 + 10,\ g(x) = 9x + 9$

$D.\quad f(x) = 9x + 9,\ g(x) = x^2 + 10$

7.

10 pts

8. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$f(x) = 7\sqrt{x + 4} - 56$

8. (a)

2 pts

(b)

2 pts

(c)

1 pts

(d)

1 pts

(e)

3 pts

(f)

1 pts

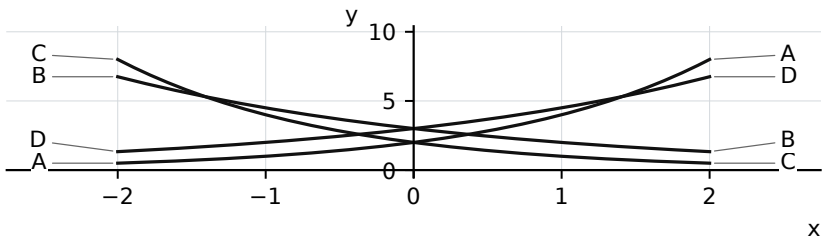
9. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 2(\frac{1}{2})^x$

$(b)\quad 2(2)^x$

$(c)\quad 3(\frac{3}{2})^x$

$(d)\quad 3(\frac{2}{3})^x$



9. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

Test continues on the other side of the sheet.

MATH 101 **36929**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Solve exactly. $8^{x-3} = 4^{x+1}$	10. 10 pts

36929

TEST PAGE 1

1.

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

2.

$x = 3,\ -\frac{1}{2}+\frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2}-\frac{1}{2}$

10 pts

3. (a)

$\mathbb{R} \setminus \{-8, 11\}$ 2 pts

(b)

$(-8,\ -\frac{1}{19})$ 2 pts

(c)

$x = 11$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-9, 0)$ 2 pts

(f)

$(0,\ -\frac{9}{11})$ 2 pts

TEST PAGE 2

4.

D

10 pts

5. (a)

$(8, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 8$ 1 pts

(d)

$(\frac{17}{2}, 0)$ 1 pts

(e)

$x \to 8^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

6. (a)

2 5 pts

(b)

$\frac{3}{2}$ 5 pts

TEST PAGE 3

7.

C

10 pts

8. (a)

$[-4, \infty)$ 2 pts

(b)

$[-56, \infty)$ 2 pts

(c)

$(60, 0)$ 1 pts

(d)

$(0,\ -42)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -4^+ : f(x) \to -56$ 1 pts

9. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

36929

TEST PAGE 4

10.

$x = 11$

10 pts

MATH 101 **64720**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

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1. Find all zeros.

$p(x) = x^3 + 5x^2 + 7x + 2$

1.

10 pts

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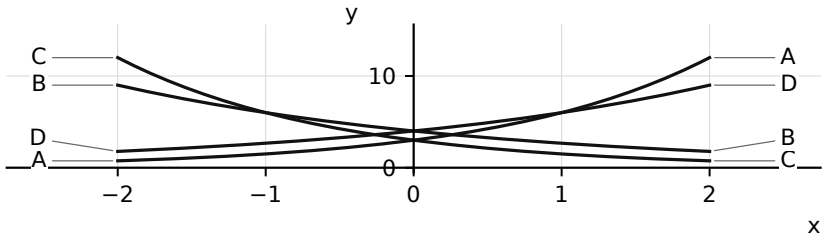
2. Match each function to graph A, B, C, or D. Each letter is used once.

$(a) \quad 3(\frac{1}{2})^x$

$(b) \quad 3(2)^x$

$(c) \quad 4(\frac{3}{2})^x$

$(d) \quad 4(\frac{2}{3})^x$



2. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

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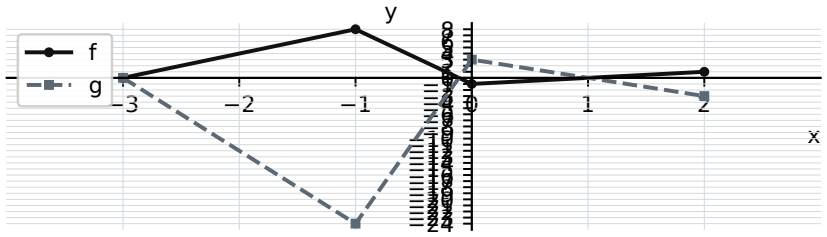
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A. \quad g(x) = f(-3x)$

$B. \quad g(x) = 3f(x)$

$C. \quad g(x) = f(x) - 3$

$D. \quad g(x) = -3f(x)$



3.

10 pts

Test continues on the other side of the sheet.

MATH 101 **64720**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x - 8) + 1$	4. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
5. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+8)(x+9)}{(x-11)(x+8)}$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts

Test continues on the next sheet.

MATH 101 **64720**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

6. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end.

$$f(x) = 7\sqrt{x+9} - 56$$

6. (a) 2 pts

(b) 2 pts

(c) 1 pts

(d) 1 pts

(e) 3 pts

(f) 1 pts

7. Evaluate exactly. Here log means base 10.

(a) $\log(0.8) + \log(1250)$

(b) $\dfrac{\log_{12}(729)}{\log_{12}(81)}$

7. (a) 5 pts

(b) 5 pts

8. Which pair satisfies h(x) = f(g(x))?

$$h(x) = (9x+9)^2 + 10$$

$A.$ $f(x) = 9x^2 + 10,\ g(x) = x + 9$

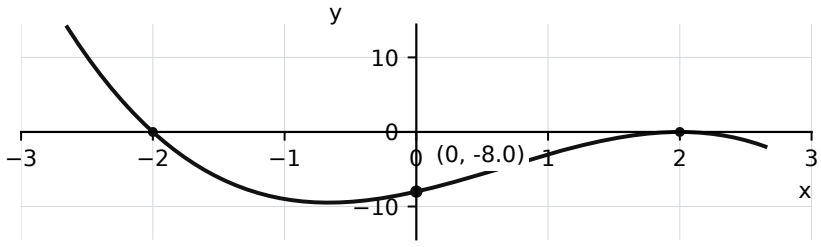
$B.$ $f(x) = x^2 + 9,\ g(x) = 9x + 10$

$C.$ $f(x) = x^2 + 10,\ g(x) = 9x + 9$

$D.$ $f(x) = 9x + 9,\ g(x) = x^2 + 10$

8. 10 pts

9. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



9. 10 pts

Test continues on the other side of the sheet.

MATH 101 **64720**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

10. Solve exactly. $\log_2(x-5) - \log_2(x-6) = 1$	10. 10 pts

64720

TEST PAGE 1

1.

$x = -2,\ -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$

10 pts

2. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

3.

D

10 pts

TEST PAGE 2

4. (a)

$(8, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 8$ 1 pts

(d)

$(\frac{25}{3}, 0)$ 1 pts

(e)

$x \to 8^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

5. (a)

$\mathbb{R} \setminus \{-8, 11\}$ 2 pts

(b)

$(-8, -\frac{1}{19})$ 2 pts

(c)

$x = 11$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-9, 0)$ 2 pts

(f)

$(0, -\frac{9}{11})$ 2 pts

TEST PAGE 3

6. (a)

$[-9, \infty)$ 2 pts

(b)

$[-56, \infty)$ 2 pts

(c)

$(55, 0)$ 1 pts

(d)

$(0, -35)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -9^+ : f(x) \to -56$ 1 pts

7. (a)

3 5 pts

(b)

$\frac{3}{2}$ 5 pts

8.

C

10 pts

9.

$-1(x+2)(x-2)^2$

10 pts

64720

TEST PAGE 4

10.

$x = 7$

10 pts

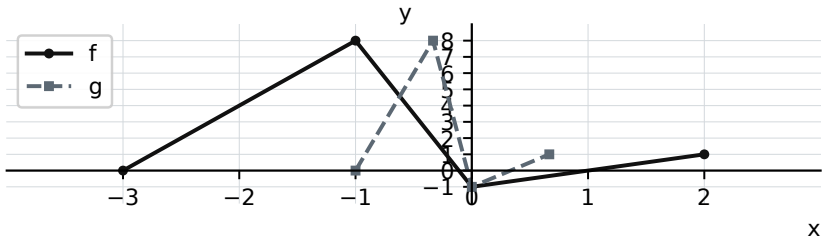
MATH 101 **83770**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Which pair satisfies h(x) = f(g(x))? $h(x) = (9x + 9)^2 + 10$ $A.\quad f(x) = 9x^2 + 10,\ g(x) = x + 9$ $B.\quad f(x) = x^2 + 9,\ g(x) = 9x + 10$ $C.\quad f(x) = x^2 + 10,\ g(x) = 9x + 9$ $D.\quad f(x) = 9x + 9,\ g(x) = x^2 + 10$	1. 10 pts
2. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.05) + \log(200000)$ $(b)\quad \frac{\log_{12}(729)}{\log_{12}(81)}$	2. (a) 5 pts (b) 5 pts
3. Find all zeros. $p(x) = x^3 + 3x^2 - 2$	3. 10 pts
4. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = 3f(x)$ $B.\quad g(x) = f(x/3)$ $C.\quad g(x) = f(x - 3)$ $D.\quad g(x) = f(3x)$ 	4. 10 pts

Test continues on the other side of the sheet.

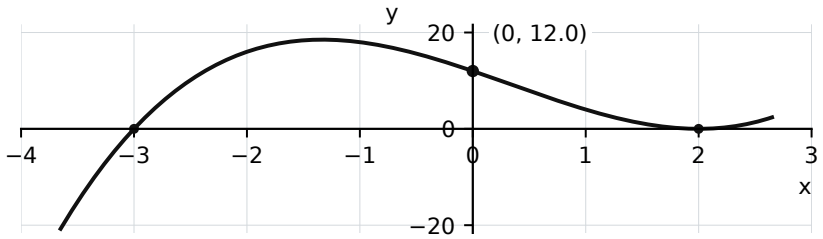
MATH 101 **83770**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

5. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x - 8) + 1$	5. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
6. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	6. 10 pts

Test continues on the next sheet.

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Diagnostic Exam

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7. Find domain; hole, vertical asymptote, and horizontal asymptote; both intercepts. Record: (a) domain; (b) hole; (c) vertical; (d) horizontal; (e) x-intercept; (f) y-intercept. $f(x) = \frac{(x+8)(x+9)}{(x-11)(x+8)}$	7. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 2 pts (f) 2 pts
8. Solve the nonlinear system. $y = x^2$ $y = 4x + 32$	8. 10 pts
9. Find domain and range; both intercepts; behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. Record: (a) domain; (b) range; (c) x-intercept; (d) y-intercept; (e) infinite end; (f) finite end. $f(x) = 7\sqrt{x+16} - 56$	9. (a) 2 pts (b) 2 pts (c) 1 pts (d) 1 pts (e) 3 pts (f) 1 pts

Test continues on the other side of the sheet.

MATH 101 **83770**

Diagnostic Exam

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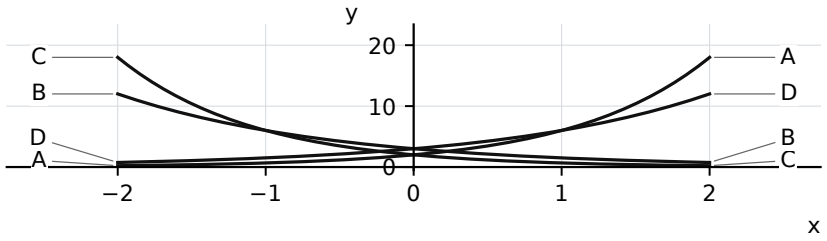
10. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(\frac{1}{3})^x$

(b) $2(3)^x$

(c) $3(2)^x$

(d) $3(\frac{1}{2})^x$



10. (a)

3 pts

(b)

2 pts

(c)

3 pts

(d)

2 pts

83770

TEST PAGE 1

1.

C

10 pts

2. (a)

4 5 pts

(b)

$\frac{3}{2}$ 5 pts

3.

$x = -1,\ -1+\sqrt{3},\ -\sqrt{3}-1$

10 pts

4.

D

10 pts

TEST PAGE 2

5. (a)

$(8, \infty)$ 2 pts

(b)

$(-\infty, \infty)$ 2 pts

(c)

$x = 8$ 1 pts

(d)

$(\frac{33}{4}, 0)$ 1 pts

(e)

$x \to 8^+ : f(x) \to -\infty$ 2 pts

(f)

$x \to \infty : f(x) \to \infty$ 2 pts

6.

$1(x+3)(x-2)^2$

10 pts

TEST PAGE 3

7. (a)

$\mathbb{R} \setminus \{-8, 11\}$ 2 pts

(b)

$(-8,\ -\frac{1}{19})$ 2 pts

(c)

$x = 11$ 1 pts

(d)

$y = 1$ 1 pts

(e)

$(-9, 0)$ 2 pts

(f)

$(0,\ -\frac{9}{11})$ 2 pts

8.

$(-4, 16); (8, 64)$

10 pts

9. (a)

$[-16, \infty)$ 2 pts

(b)

$[-56, \infty)$ 2 pts

(c)

$(48, 0)$ 1 pts

(d)

$(0,\ -28)$ 1 pts

(e)

$x \to \infty : f(x) \to \infty$ 3 pts

(f)

$x \to -16^+ : f(x) \to -56$ 1 pts

83770

TEST PAGE 4

10. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts